

CAF120 (P53836) — Evaluation of "protein kinase activity" (GO:0004672)

Gene: CAF120 / YNL278W · *Saccharomyces cerevisiae* (NCBITaxon:559292) **UniProt:** P53836 (CA120_YEAST), 1060 aa **Focus type:** function_assignment — does the gene product directly have this molecular function? **Hypothesis under review:** CAF120 has protein kinase activity (GO:0004672) **Current annotation:** GO:0004672 · evidence IBA · reference GO_REF:0000033 (GO_Central phylogenetic annotation)

Summary

The seed hypothesis that CAF120 possesses **protein kinase activity (GO:0004672)** is **refuted**. Three independent lines of evidence converge: (1) programmatic domain analysis shows P53836 carries **only a Pleckstrin-homology (PH) domain** and its fungal derivative — there is no protein-kinase domain of any kind; (2) a direct scan of the 1060-residue sequence finds **none** of the conserved eukaryotic protein-kinase (ePK) catalytic motifs required for phosphotransfer; and (3) the experimentally supported function of Caf120 is regulatory/adaptor in nature — a meiotic crossover regulator physically linked to the Exo1–MutLy recombination machinery and, per UniProt curation, associated with the CCR4–NOT context — with no report of catalytic activity.

The GO:0004672 annotation is an **IBA phylogenetic mis-propagation**. It is co-annotated with GO:0007165 (signal transduction, also IBA) in the same record, and PH domains are shared with genuine kinases (e.g., AKT/PKB-type AGC kinases carry a PH module adjacent to their catalytic domain). The most parsimonious interpretation is that the GO_Central pipeline propagated "protein kinase activity + signal transduction" from a PH-domain-containing kinase clade to CAF120, which possesses the PH module but not the kinase domain.

The recommended curation action is to **remove** GO:0004672 from CAF120 — not to retain it, generalize it to a broader transferase/kinase term, or replace it with a more specific kinase term. This judgment is a structural-impossibility argument (the catalytic residues are

physically absent) rather than merely "not yet tested," which makes the refutation strong. The residual uncertainty is limited to whether some wholly non-canonical catalytic mechanism could exist, for which there is no evidence.

Key Findings

Finding 1 — CAF120 has a PH-domain architecture with no kinase domain

Domain analysis via the InterPro API (queried this run) returns a single, internally consistent picture: **P53836 is a Pleckstrin-homology (PH) domain protein**. The matched signatures include Pfam **PF00169** (PH domain) and **PF25381** (fungal PH-like), PROSITE **PS50003** (PH profile), SMART **SM00233** (PH), Gene3D **2.30.29.30** (PH-domain-like β -sandwich fold), the InterPro PH-like superfamily **IPR011993**, and the dedicated fungal family **IPR058155** ("**Skg3/CAF120-like PH domain**").

Critically, **no kinase signatures are present at all**. There is no Pfam PF00069 (Pkinase), no InterPro IPR000719 (protein kinase domain), and no PANTHER protein-kinase family assignment. The PH-like β -sandwich fold is structurally unrelated to the bilobal α/β protein-kinase fold. A PH domain is a lipid/protein-interaction module that recruits proteins to membranes or partners; it confers no catalytic activity by itself. The existence of a *dedicated* fungal family term (IPR058155) built around the PH module — not a catalytic domain — indicates the correct family placement for CAF120 is a PH-adaptor family, not a kinase family.

Finding 2 — The eukaryotic protein-kinase catalytic motifs are absent from the sequence

A direct motif scan of the full 1060-aa P53836 sequence tested each conserved catalytic element of the canonical ePK domain. All were absent:

Catalytic element	Consensus	Function	Present in P53836?
Glycine-rich loop (P-loop)	G-x-G-x-x-G	Positions ATP phosphates	No
β 3 lysine (VAIK)	V-A-I-K	Anchors α/β phosphates of ATP	No
Catalytic loop	H-R-D	Catalytic aspartate (proton acceptor)	No
Mg-binding loop	D-F-G	Coordinates Mg^{2+}	Only stray "DFG" tripeptides in N-terminal/disordered region, not in a catalytic domain
PROSITE PS00107	ATP-binding kinase signature	ATP binding	No match
PROSITE PS00108	Ser/Thr kinase active-site signature	Catalysis	No match

The simultaneous absence of the P-loop, the VAIK lysine, and the HRD aspartate is decisive. These form the minimal catalytic core: the P-loop and lysine bind and orient ATP, while the HRD aspartate is the catalytic base that deprotonates the substrate hydroxyl. Even pseudokinases — catalytically dead but retaining the fold — usually preserve recognizable remnants of these motifs. CAF120 has none of the scaffold ordered within a ~250–300-aa domain, indicating it does not belong to the protein-kinase superfamily at all. The two "DFG" tripeptides that occur by chance fall in the N-terminus and a disordered region rather than within any structured catalytic module, so they are not evidence of a kinase.

Finding 3 — The experimentally supported role of Caf120 is regulatory, not catalytic

Two primary studies characterize Caf120, and neither reports or implies catalytic kinase activity:

- **Meiotic crossover regulation.** In a proteomic dissection of homologous-recombination enzyme networks, Caf120 was identified among the meiosis-specific interactors of the MutLy–Exo1 crossover machinery and classified as a **regulator of crossing-over** (PMID: 31351878). This places Caf120 in a physical/regulatory relationship with the recombination apparatus — an adaptor role, not an enzymatic one.
- **Stress-dependent relocalization.** In a proteome-wide survey of protein location and function under stress, Caf120 was one of several proteins that **change subcellular location under DNA-damaging conditions** (PMID: 25028499). Regulated relocalization is consistent with a PH-domain adaptor redeployed in response to signals; no kinase activity is described.

UniProt further curates CAF120 within the CCR4-NOT context (transcription regulation / mRNA deadenylation) and notes it "Belongs to the CAF120 family." A "Phosphoprotein" keyword marks CAF120 as a kinase **substrate/target**, not a kinase — a common source of confusion, but the opposite of the hypothesis. None of these curated or experimental descriptions invokes phosphotransfer chemistry.

Mechanistic Model / Interpretation

The evidence assembles into a clear model of how a spurious kinase annotation arose and why it should be removed.

WHAT CAF120 ACTUALLY IS

1060-aa protein

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|
[ PH domain  (PF00169) ]
[ fungal Skg3/CAF120 PH ]
[ IPR058155 family   ]

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|
lipid / protein binding
adaptor & recruitment

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regulatory / substrate roles:
- meiotic crossover regulation
  (Exo1/MutLgamma, <a href="https://pubmed.ncbi.nlm.nih.gov/31351878/" rel="noopener noreferrer" title="Visi
- stress relocalization
  (<a href="https://pubmed.ncbi.nlm.nih.gov/25028499/" rel="noopener noreferrer" title="Visi
- CCR4-NOT context (UniProt)
- phosphoprotein = KINASE TARGET,
  not kinase

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WHAT THE IBA CLAIMS

"protein kinase activity"

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|
requires bilobal ePK fold

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requires: GxGxxG P-loop
          VAIK Lys
          HRD Asp
          DFG motif

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|
NONE present in a catalytic
domain in P53836

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=> structurally impossible

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How the mis-annotation propagated. GO:0004672 is annotated IBA (Inferred from Biological Ancestor) via GO_REF:0000033, the GO_Central phylogenetic pipeline, in which an experimentally supported annotation on one family member is propagated to uncharacterized members in the same inferred ancestral clade. PH domains occur in many signaling proteins, and a subset of *bona fide* kinases — notably the AGC-family AKT/PKB — carry a PH module adjacent to their kinase domain. If the reference tree grouped CAF120's PH module with the PH modules of such kinases, "protein kinase activity" and "signal transduction" leaf annotations could propagate to CAF120 even though it has *only* the PH module. The co-occurrence of GO:0007165 (signal transduction) IBA in the same record is the fingerprint of exactly this PH-domain-driven mis-grouping.

The correct molecular picture. CAF120 is a non-catalytic PH-domain adaptor. Its PH β -sandwich most plausibly mediates membrane or protein interactions that position CAF120 within regulatory assemblies — the meiotic recombination network and the CCR4-NOT context. Its function is to *localize and connect*, not to *phosphorylate*.

Evidence Base

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence/ limitation
InterPro API (this run; IPR011993, IPR058155; Pfam PF00169, PF25381; PROSITE PS50003; SMART SM00233; Gene3D 2.30.29.30)	Structural/ evolutionary (domain)	Refutes	Does CAF120 contain a protein-kinase domain?	Only PH domain + PH-like superfamily + fungal "Skg3/CAF120-like PH"; no kinase domain (no PF00069, no IPR000719, no PANTHER kinase family)	<i>S. cerevisiae</i> P53836 sequence	High. Absence of a kinase hit is definitive for the ePK family
Sequence motif scan (this run)	Computational	Refutes	Are ePK catalytic motifs present?	None found: no GxGxxG P-loop, no VAIK Lys, no HRD, no PS00107/PS00108. Stray "DFG" tripeptides fall in N-terminal/disordered region, not a catalytic domain	1060-aa sequence	High. Genuine ePKs require the motifs ordered within a ~250–300-aa domain
UniProt P53836 (Swiss-Prot, reviewed)	Review/ database	Competing (different function)	What is CAF120's function?	CCR4-NOT context (transcription regulation / mRNA deadenylation); "Belongs to the CAF120 family"; keyword Phosphoprotein	<i>S. cerevisiae</i>	High (manually curated). Function is not catalytic

Citation	Evidence type	Supports/ Refutes	Claim tested	Key finding	Context	Confidence limitation
				(substrate, not enzyme)		
PMID: 31351878	Mutant/ interaction (primary)	Competing	Native role of Caf120?	Identified as a regulator of meiotic crossing-over , a meiosis- specific interactor of the MutLy- Exo1 recombination machinery	Yeast meiosis	Medium-h No kinase activity implied
PMID: 25028499	Localization (primary)	Qualifies	Condition- dependent behavior?	Caf120 changes subcellular location under DNA damage	Yeast stress proteomics	Medium. Regulatory relocaliza not enzym
GO:0004672 IBA + GO:0007165 IBA (GO_Central, GO_REF:0000033)	Computational (phylogenetic)	Source of hypothesis	Origin of kinase annotation	Both terms IBA from the same reference; kinase+signal- transduction pairing on a PH-only protein is a classic mis- propagation signature	GO_Central	The IBA is artefact b evaluated independe support

GO Curation Implications

Lead (requires curator verification): Remove GO:0004672 (protein kinase activity, MF) from CAF120. The IBA lacks any catalytic-domain basis and conflicts with the curated molecular identity of the protein. Do **not** replace it with any other kinase/transferase MF term.

Reasoning by GO aspect:

- **Molecular Function (MF).** GO:0004672 is unsupported and contradicted by domain and motif analysis. If a positive MF term is desired, the only defensible option is a **PH-domain-based binding function** (lipid/phosphoinositide binding), and only if curator inspection of the PH ligand specificity supports it; otherwise leave MF unannotated rather than assert catalysis. Avoid the uninformative "protein binding" as a terminal recommendation.
- **Biological Process (BP).** The co-propagated GO:0007165 (signal transduction) IBA warrants the same scrutiny and is likely another mis-propagation. More defensible BP leads reflect the experimental literature — meiotic recombination / regulation of reciprocal meiotic recombination ([PMID: 31351878](#)) — flagged for curator verification.
- **Cellular Component (CC).** Stress-dependent relocalization ([PMID: 25028499](#)) and CCR4-NOT context (UniProt) are the relevant CC leads, not anything kinase-related.

GO decision table:

Term	Aspect	Current	Recommended action	Basis
GO:0004672 protein kinase activity	MF	IBA (GO_REF:0000033)	Remove	No kinase domain; no catalytic residues
GO:0007165 signal transduction	BP	IBA (co-annotation)	Flag / likely remove	Same PH-domain mis-propagation signature
Meiotic recombination regulation	BP	—	Candidate add (curator verify)	P 31351878

Mechanistic Scope

The molecular function under test is **catalytic transfer of a phosphate from ATP to a protein substrate** (phosphotransfer). This requires a spatially organized active site — a P-loop and $\beta 3$ lysine to bind/orient ATP, an HRD aspartate as catalytic base, and a DFG aspartate to coordinate Mg^{2+} . CAF120 lacks every one of these, so the *direct gene-product activity* hypothesized is physically precluded.

What CAF120 *does* have is a PH domain, whose direct molecular activity is **binding** (phosphoinositide lipids and/or protein partners) — a recruitment/scaffolding activity. Everything else attributed to CAF120 — regulation of meiotic crossovers, stress relocalization, CCR4-NOT association, and its status as a phosphoprotein — is either a **downstream, context-dependent regulatory role** mediated through interactions, or evidence that CAF120 is a kinase *target*. The kinase hypothesis conflates "participates in signaling/regulatory processes" (plausible for a PH adaptor) with "is itself a catalytic kinase" (unsupported and structurally impossible).

Conflicts and Alternatives

- **Paralog / family confusion (most likely cause).** PH domains are present in many signaling kinases (AKT/PKB, PDK1, GRKs). A phylogenetic annotation propagated "protein kinase activity" down a reference tree from a PH-domain-containing kinase clade to CAF120, which shares only the non-catalytic PH module. This is the classic mechanism of IBA over-annotation.
 - **Substrate-vs-enzyme confusion.** CAF120 is a phosphoprotein (target of kinases), not a kinase; the "Phosphoprotein" keyword can be misread as implying enzymatic activity.
 - **Co-annotation artifact.** GO:0007165 appearing alongside GO:0004672 is itself evidence of a bundled mis-propagation rather than two independent, evidence-based calls.
 - **No competing experimental claim.** There is no primary paper asserting kinase activity for CAF120; the annotation rests solely on the IBA inference, making the conflict one-sided.
 - **Atypical/divergent kinase possibility (considered and rejected).** Highly diverged atypical kinases still adopt a nucleotide-binding kinase-like fold detectable by profile HMMs; InterPro/Pfam searches returned no kinase-related hit of any kind.
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Exact PANTHER/PAINT node that seeded the IBA	InterPro PANTHER endpoint returned no record this run	Confirms the propagation source and lets curators fix the tree annotation	Query GO_Central/PANTHER PAINT tree for the P53836 leaf and its annotated ancestral node
No direct in-vitro kinase assay exists for CAF120	PubMed searches found only CCR4-NOT / meiosis / stress roles	A direct negative assay would be definitive, though absence of catalytic residues already predicts no activity	In-vitro autophosphorylation/ substrate assay (predicted negative)
AlphaFold fold confirmation of non-PH regions	AF model exists (P53836); much of the sequence is predicted disordered	Rules out a cryptic kinase fold	Inspect AlphaFold pLDDT/ domain segmentation; expect PH domain + long disordered tail, no kinase lobe
PH-domain ligand specificity	Domain identity clear, but specific ligand not determined here	A positive MF term depends on it	Structural docking / lipid-overlay (PIP-strip) assays

Additional limitation: the refutation is primarily bioinformatic (domain + motif) plus the absence of any positive experimental kinase report. There is no wet-lab negative kinase assay in the literature; however, the physical absence of the catalytic residues provides a strong mechanism-based refutation, not merely an evidential one.

Discriminating Tests

1. **PANTHER tree inspection (fastest).** Locate P53836 in its GO_Central PANT tree and check whether the kinase MF was assigned at an ancestral node dominated by PH-domain kinases — confirms mis-propagation and gives curators the exact node to correct.
 2. **HMMER/hmmscan against Pfam PF00069 (Pkinase / PK_Tyr_Ser-Thr).** Expected: no hit (corroborates the InterPro result with an independent tool).
 3. **Structural superposition** of the CAF120 AlphaFold model onto a canonical kinase (e.g., PKA) catalytic domain. Expected: no bilobal match beyond incidental PH-fold.
 4. **PH-domain ligand assay.** Lipid-overlay / liposome-binding to determine phosphoinositide specificity, supporting a correct positive binding MF term.
 5. **(Definitive, low-priority) in-vitro kinase assay** with generic substrate + [γ - 32 P]ATP. Predicted: negative.
 6. **Family-wide IBA audit.** Pull all IPR058155 (Skg3/CAF120-like PH) members and check for GO:0004672 IBA to identify and correct systematic over-annotation.
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Proposed Follow-up Actions / Curation Leads (require curator verification)

- **Action:** Flag GO:0004672 (protein kinase activity, IBA, GO_REF:0000033) as an over-annotation for CAF120 and remove it from the gene review; do not replace with any other kinase MF term.
- **Also review (out of focus):** GO:0007165 (signal transduction, IBA) as the co-propagated term.
- **Candidate supporting references / snippets to verify:**
 - [PMID: 31351878](#) — "*Functional analyses of meiosis-specific interactors of MutLy-Exo1 identified Rtk1, Caf120, and Chd1 as regulators of crossing-over.*" (establishes a real, non-kinase function).
 - [PMID: 25028499](#) — "*under DNA-damaging conditions, Tsr1, Caf120, Dip5, Skg6, Lte1, and Nnf2 change subcellular location.*" (regulatory relocation, no catalytic activity).
 - UniProt P53836 FUNCTION/SIMILARITY blocks — "Belongs to the CAF120 family."

- **Suggested curator question:** Was the kinase MF asserted at a PH-domain-containing ancestral node in the PANTHER tree? If so, correct the ancestral annotation to stop propagation.
- **Suggested experiments (if any doubt remains):** in-vitro kinase assay (predicted negative); AlphaFold fold comparison to a kinase; PH-domain lipid-binding assay (to assign a correct positive MF).

Provenance

Computational checks executed during this investigation: - UniProt REST fetch of P53836 (function, features, GO, cross-references, sequence). - InterPro API domain enumeration for P53836 (PH-only domain architecture; PF00169, PF25381, PS50003, SM00233, Gene3D 2.30.29.30, IPR011993, IPR058155; no kinase family). - Regex/PROSITE scan of the 1060-aa sequence for ePK catalytic motifs (GxGxxG, VAIK, HRD, DFG, PS00107, PS00108) — all negative within a catalytic domain.

Conclusion

CAF120 (P53836) is a Pleckstrin-homology-domain adaptor protein of the fungal Skg3/CAF120-like family, with experimentally supported regulatory roles in meiotic crossover control and stress-responsive relocalization, and a database-curated association with the CCR4-NOT context. It has no protein-kinase domain and lacks every conserved catalytic residue required for phosphotransfer. The GO:0004672 (protein kinase activity) IBA annotation is a PH-domain-driven phylogenetic over-annotation and should be **removed** rather than retained, generalized, or replaced with another catalytic term.